ALYSSA MATICKA

647-544-6477 | alyssa.matic@gmail.com | LinkedIn | Ajax, Ontario | Kingston, Ontario

PROFILE

Mechanical Engineering student with hands-on experience designing, building, and commissioning a complete solar thermal heating system. Strong foundation in heat transfer, thermodynamics, fluid mechanics, and mechanical design. Passionate about HVAC systems, sustainable infrastructure, and renewable energy. Eager to contribute to engineering teams focused on building performance, clean energy, and low-carbon systems design.

EDUCATION

Bachelor of Applied Science – Mechanical Engineering Expected April 2028
Queen's University, Kingston, ON | *Dean's Scholar Award (2024–2025)* | *GPA: 3.9*
Relevant Coursework: Thermodynamics, Fluid Dynamics, Heat Transfer, Engineering Design, Python, MATLAB, C++

TECHNICAL SKILLS

Thermal & Mechanical: Heat transfer analysis (conduction, convection, radiation), HVAC system sizing, piping design, fluid mechanics, thermodynamic cycle analysis, structural analysis

CAD & Simulation: SolidWorks (3D modelling, assemblies, engineering drawings, FEA/stress analysis)

Programming & Data: Python (data processing, automation, firmware), MATLAB, Arduino (sensor integration, control systems), Excel

Field & Technical: Hands-on system installation (solar collectors, copper piping, valves, baseboard heating), site coordination, WHMIS certified

Documentation: Engineering reports, design memos, calculation packages, compliance documentation, stakeholder communication

PROJECT EXPERIENCE

Executive Project Manager – Solar Thermal Heating System | *Queen's Solar Design Team* Sept 2025 – Present

- Performed heat transfer calculations to size an evacuated tube solar thermal space heating system; designed full infrastructure from collectors through to baseboard delivery
- Physically built and commissioned the complete system: assembled vacuum tube solar collectors, fabricated and soldered copper piping, installed valves, pump, expansion tank, and baseboard heating
- Collected and analyzed thermal performance data; produced technical documentation including calculation packages, design assumptions, and project schedules
- Managed team coordination and stakeholder communications across a multi-semester project, delivering all milestones on time

Project Lead – Nuclear Waste Management System | *Alberta SMR Conceptual Design* Jan 2026 – Present

- Led technical design and compliance documentation for a 3,552 m^3/year low-level nuclear waste system applying CNSC and Transport Canada regulatory frameworks
- Produced structured engineering reports assessing economic and environmental viability of design decisions, including feasibility analysis and alternatives evaluation

Mechanical Lead – Automated Water Pre-Treatment System | *Queen's Engineering Design* Sept – Dec 2024

- Designed a mechanical mixing assembly and full system model in SolidWorks, including FEA validation for structural performance
- Programmed an Arduino-based control system with sensor feedback for automated process control; applied fluid mechanics and iterative testing to validate performance

- Produced engineering drawings, test procedures, and design documentation through the full deliverable cycle

Mechanical Subteam Member | *Queen's Relectric Car Team* Sept 2025 – Present

- Designed a custom motor mount for a Micro 841 electric motor in SolidWorks; produced 3D models, assemblies, and engineering drawings validated through static and frequency analysis
- Applied iterative design review to confirm structural performance under dynamic load conditions; attending Purdue Grand Prix as pit crew, April 2026

Analysis & Report Coordinator – Lunar Robotic Harvester (HarveStar) | *Canadian Space Agency (OASYS)* Jan – Apr 2025

- Led structural and mechanical analysis on a robotic end-effector; coordinated all technical reporting across the team
- Wrote Python (CircuitPython) firmware for a 4-DOF robotic arm with inverse kinematics and 18-zone joint constraint safety system

LEADERSHIP & WORK EXPERIENCE

Founder & Swim Instructor | *Alyssa's Aquatics* May 2025 – Present

- Founded and independently operated a swim instruction business; managed scheduling, client communications, finances, and a custom booking website

Lifeguard, Swim Instructor & Bus Counsellor | *Camp Robin Hood / Town of Ajax* Jul 2021 – Aug 2025

- Administered first aid; trained and mentored new staff; maintained safety records and coordinated emergency procedures

EXTRACURRICULARS & INTERESTS

- Queen's Solar Design Team – Executive Project Manager (2025–present)
- Queen's Relectric Car Team – Mechanical Subteam (2025–present)
- Queen's Nuclear Engineering Team – Sponsorship Director (2026–present)
- Canadian Nuclear Society – Student Member (2026–present)
- Queen's University Sustainable Solutions & Technology Competition – Speakers Director (2026–present)
- Personal Project: Designing and building a solar-powered RC car — independent electrical and mechanical systems design
- Competitive Athletics: CSL Soccer (Whitby FC); Rugby Team Captain (2020–2024)